6

Conditional Distributions

Building on Section 1.10, this chapter provides a more thorough and proper treatment of conditioning. Section 6.4 gives a proof of the factorization theorem (Theorem 3.6).

6.1 Joint and Marginal Densities

Let X be a random vector in $\mathbb{R}^m$, let Y be a random vector in $\mathbb{R}^n$, and let $Z = (X, Y)$ in $\mathbb{R}^{m+n}$. Suppose P_Z has density p_Z with respect to $\mu \times \nu$ where μ and ν are measures on $\mathbb{R}^m$ and $\mathbb{R}^n$. This density p_Z is called the *joint density* of X and Y. Then

$$P(Z \in B) = \iint 1_B(x, y) p_Z(x, y) \, d\mu(x) \, d\nu(y).$$

By Fubini's theorem, the order of integration here can be reversed. To compute $P(X \in A)$ from this formula, note that $X \in A$ if and only if $Z \in A \times \mathbb{R}^n$. Then because $1_{A \times \mathbb{R}^n}(x, y) = 1_A(x)$,

$$\begin{aligned} P(X \in A) = P(Z \in A \times \mathbb{R}^n) &= \iint 1_A(x) p_Z(x, y) \, d\nu(y) \, d\mu(x) \\ &= \int_A \left\{ \int p_Z(x, y) \, d\nu(y) \right\} d\mu(x). \end{aligned}$$

From this, X has density

$$p_X(x) = \int p_Z(x, y) \, d\nu(y) \tag{6.1}$$

with respect to μ. This density p_X is called the *marginal density* of X. Similarly, Y has density

$$p_Y(y) = \int p_Z(x, y) \, d\mu(x),$$

called the marginal density of Y.

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Example 6.1. Suppose μ is counting measure on $\{0, 1, \dots, k\}$ and ν is Lebesgue measure on $\mathbb{R}$. Define

$$p_Z(x, y) = \begin{cases} \binom{k}{x} y^x (1-y)^{k-x}, & x = 0, 1, \dots, k,\ y \in (0, 1); \\ 0, & \text{otherwise.} \end{cases}$$

By (6.1), X has density

$$p_X(x) = \int_0^1 \binom{k}{x} y^x (1-y)^{k-x} dy = \frac{1}{k+1}, \qquad x = 0, 1, \dots, k.$$

(The identity $\int_0^1 u^{\alpha-1}(1-u)^{\beta-1}\,du = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$ is used to evaluate the integral.) This is the density for the uniform distribution on $\{0, 1, \dots, k\}$. To find the marginal density of Y we integrate the joint density against μ. For $y \in (0, 1)$,

$$p_Y(y) = \int p_Z(x, y)\,d\mu(x) = \sum_{x=0}^{k} \binom{k}{x} y^x (1-y)^{k-x} = 1;$$

and if $y \notin (0, 1)$, $p_Y(y) = 0$. Therefore Y is uniformly distributed on $(0, 1)$.

6.2 Conditional Distributions

Let X and Y be random vectors. The definition of the conditional distribution Q_x of Y given $X = x$ is related to our fundamental smoothing identity. Specifically, if $E|f(X, Y)| < \infty$, we should have

$$Ef(X, Y) = EE[f(X, Y)|X], \tag{6.2}$$

with $E[f(X, Y)|X] = H(X)$ and

$$H(x) = E[f(X, Y)|X = x] = \int f(x, y)\,dQ_x(y).$$

Written out, (6.2) becomes

$$Ef(X, Y) = \int H(x)\,dP_X(x) = \iint f(x, y)\,dQ_x(y)\,dP_X(x). \tag{6.3}$$

The formal definition requires that (6.3) holds when f is an indicator of $A \times B$. Then (6.2) or (6.3) will hold for general measurable f provided $E|f(X, Y)| < \infty$.

Definition 6.2. *The function Q is a* conditional distribution *for Y given X, written*

$$Y|X = x \sim Q_x,$$

if

1. *$Q_x(\cdot)$ is a probability measure for all x,*
2. *$Q_x(B)$ is a measurable function of x for any Borel set B, and*
3. *for any Borel sets A and B,*

$$P(X \in A, Y \in B) = \int_A Q_x(B)\, dP_X(x).$$

When X and Y are random vectors, conditional distributions will always exist.[1] Conditional probabilities can be defined in more general settings, but assignments so that $Q_x(\cdot)$ is a probability measure may not be possible.

The stated definition of conditional distributions is not constructive. In the setting of Section 6.1 in which X and Y have joint density p_Z with respect to $\mu \times \nu$, conditional distributions can be obtained explicitly using the following result.

Theorem 6.3. *Suppose X and Y are random vectors with joint density p_Z with respect to $\mu \times \nu$. Let p_X be the marginal density for X given in* (6.1)*, and let $E = \{x : p_X(x) > 0\}$. For $x \in E$, define*

$$p_{Y|X}(y|x) = \frac{p_Z(x, y)}{p_X(x)}, \tag{6.4}$$

and let Q_x be the probability measure with density $p_{Y|X}(\cdot|x)$ with respect to ν. When $x \notin E$, define $p_{Y|X}(y|x) = p_0(y)$, where p_0 is the density for an arbitrary fixed probability distribution P_0, and let $Q_x = P_0$. Then Q is a conditional distribution for Y given X.

Proof. Part one of the definition is apparent, and part two follows from measurability of p_Z. It is convenient to establish (6.3) directly; part three of the definition then follows immediately. First note that $P(X \in E) = 1$, and without loss of generality we can assume that $p_Z(x, y) = 0$ whenever $x \notin E$. (If not, just change $p_Z(x, y)$ to $p_Z(x, y)1_E(x)$—these functions agree for a.e. (x, y) $(\mu \times \nu)$, and either can serve as the joint density.) Then $p_Z(x, y) = p_X(x)p_{Y|X}(y|x)$ for all x and y, and the right-hand side of (6.3) equals

$$\iint f(x, y)p_{Y|X}(y|x)\, d\nu(y)\, p_X(x)\, d\mu(x)$$
$$= \iint f(x, y)p_Z(x, y)\, d\nu(y)\, d\mu(x) = Ef(X, Y).$$

$\square$

When X and Y are independent, $p_Z(x, y) = p_X(x)p_Y(y)$, and so

[1] When X is a random variable, this is given as Theorem 33.3 of Billingsley (1995). See Chapter 5 of Rao (2005) for more general cases.

$$p_{Y|X}(y|x) = \frac{p_X(x)p_Y(y)}{p_X(x)} = p_Y(y),$$

for $x \in E$. So the conditional and marginal distributions for Y are the same (for a.e. x).

Example 6.1, continued. Because $p_Y(y) = 1$, $y \in (0, 1)$,

$$p_{X|Y}(x|y) = p_Z(x, y) = \binom{k}{x} y^x (1-y)^{k-x}, \qquad x = 0, 1, \dots, k.$$

As a function of x with y fixed, this is the mass function for the binomial distribution with success probability y and k trials. So

$$X|Y = y \sim \text{Binomial}(k, y). \tag{6.5}$$

Similarly, recalling that $p_X(x) = 1/(k+1)$,

$$\begin{aligned} p_{Y|X}(y|x) &= \frac{p_Z(x,y)}{p_X(x)} = (k+1)\binom{k}{x} y^x (1-y)^{k-x} \\ &= \frac{\Gamma(k+2)}{\Gamma(x+1)\Gamma(k-x+1)} y^{x+1-1}(1-y)^{k-x+1-1}, \qquad y \in (0,1). \end{aligned}$$

This is the density for the beta distribution, and so

$$Y|X = x \sim \text{Beta}(x+1, k-x+1).$$

To illustrate how smoothing might be used to calculate expectations in this example, as the binomial distribution in (6.5) has mean ky,

$$E[X|Y] = kY.$$

So, by smoothing,

$$EX = EE[X|Y] = kEY = k\int_0^1 y\,dy = \frac{k}{2}.$$

Summation against the mass function for X gives the same answer:

$$EX = \sum_{x=0}^{k} \frac{x}{k+1} = \frac{k}{2}.$$

To compute EX^2 using smoothing, because the binomial distribution in (6.5) has second moment $ky(1-y) + k^2y^2$,

$$\begin{aligned} EX^2 &= EE[X^2|Y] = E\big[kY(1-Y) + k^2Y^2\big] \\ &= \int_0^1 \big(ky(1-y) + k^2y^2\big)\,dy = \frac{k(1+2k)}{6}. \end{aligned}$$

Summation against the mass function for X gives

$$EX^2 = \sum_{x=0}^{k} \frac{x^2}{k+1},$$

so these calculations show indirectly that

$$\sum_{x=0}^{k} x^2 = \frac{k(k+1)(2k+1)}{6}.$$

(This can also be proved by induction.)

6.3 Building Models

To develop realistic models for two or more random vectors, it is often convenient to specify a joint density, using (6.4), as

$$p_Z(x,y) = p_X(x)p_{Y|X}(y|x).$$

The thought process using this equation would involve first choosing a marginal distribution for X and then combining this marginal distribution with a suitable distribution for Y if X were known. This equation can be extended to several vectors. If $p(x_k|x_1,\dots,x_{k-1})$ denotes the conditional density of X_k given $X_1 = x_1,\dots,X_{k-1} = x_{k-1}$, then the joint density of $X_1,\dots,X_n$ is

$$p_{X_1}(x_1)p(x_2|x_1)\cdots p(x_n|x_1,\dots,x_{n-1}). \tag{6.6}$$

Example 6.4. Models for Time Series. Statistical applications in which variables are observed over time are widespread and diverse. Examples include prices of stocks, measurements of parts from a production process, or growth curve data specifying size or dimension of a person or organism over time. In most of these applications it is natural to suspect that the observations will be dependent. For instance, if X_k is the log of a stock price, a model with

$$X_k|X_1 = x_1,\dots,X_{k-1} = x_{k-1} \sim N(x_{k-1}+\mu,\sigma^2)$$

may be natural. If $X_1 \sim N(x_0+\mu,\sigma^2)$, then by (6.6), $X_1,\dots,X_n$ will have joint density

$$\prod_{k=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x_k - x_{k-1} - \mu)^2}{2\sigma^2}\right].$$

Differences $X_k - X_{k-1}$ here are i.i.d. from $N(\mu,\sigma^2)$, and the model here for the joint distribution is called a *random walk.*

Another model for variables that behave in a more stationary fashion over time might have

$$X_k|X_1 = x_1, \dots, X_{k-1} = x_{k-1} \sim N(\rho x_{k-1}, \sigma^2),$$

where $|\rho| < 1$. If $X_1 \sim N(\rho x_0, \sigma^2)$, then by (6.6) the joint density is

$$\prod_{k=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x_k - \rho x_{k-1})^2}{2\sigma^2}\right].$$

This is called an *autoregressive* model.

Example 6.5. A Simple Model for Epidemics. For any degree of realism, statistical models for epidemics must allow substantial dependence over time, and conditioning arguments can be quite useful in attempts to incorporate this dependence in a natural fashion. To illustrate, let us develop a simple model based on suspect assumptions. Improvements with more realistic assumptions should give practical and useful models

Let N denote the size of the population of interest, and let X_i denote the number of infected individuals in the population at stage i. Assume that once someone is infected, they stay infected. Also, assume that the chance an infected individual infects a noninfected individual during the time interval between two stages is $p = 1 - q$ and that all chances for infection are independent. Then the chance a noninfected person stays noninfected during the time interval between stages k and $k+1$, given $X_k = x_k$ (and other information about the past), is q^{x_k}, and so the number of people newly infected during this time interval, $X_{k+1} - X_k$, will have a binomial distribution. Specifically

$$X_{k+1} - X_k|X_1 = x_1, \dots, X_k = x_k \sim \text{Binomial}(N - x_k, 1 - q^{x_k}).$$

This leads to conditional densities (mass functions)

$$p(x_{k+1}|x_1, \dots, x_k) = \binom{N - x_k}{x_{k+1} - x_k} \bigl(1 - q^{x_k}\bigr)^{x_{k+1} - x_k} \bigl(q^{x_k}\bigr)^{N - x_{k+1}}.$$

The product of these gives the joint mass function for $X_1, \dots, X_n$.

6.4 Proof of the Factorization Theorem[2]

To prove the factorization theorem (Theorem 3.6) we need to work directly from the definition of conditional distributions, for in most cases T and X will not have a joint density with respect to any product measure. To begin, suppose P_θ, $\theta \in \Omega$, has density

$$p_\theta(x) = g_\theta\bigl(T(x)\bigr)h(x) \tag{6.7}$$

[2] This section is optional; the proof is fairly technical.

with respect to μ. Modifying h, we can assume without loss of generality that μ is a probability measure equivalent[3] to the family $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$. Let E^* and P^* denote expectation and probability when $X \sim \mu$; let G^* and G_θ denote marginal distributions for $T(X)$ when $X \sim \mu$ and $X \sim P_\theta$; and let Q be the conditional distribution for X given T when $X \sim \mu$. To find densities for T,

$$\begin{aligned} E_\theta f(T) &= \int f\bigl(T(x)\bigr) g_\theta\bigl(T(x)\bigr) h(x)\, d\mu(x) \\ &= E^* f(T) g_\theta(T) h(X) \\ &= \iint f(t) g_\theta(t) h(x)\, dQ_t(x)\, dG^*(t) \\ &= \int f(t) g_\theta(t) w(t)\, dG^*(t), \end{aligned}$$

where

$$w(t) = \int h(x)\, dQ_t(x).$$

If f is an indicator function, this shows that G_θ has density $g_\theta(t)w(t)$ with respect to G^*. Next, define $\tilde{Q}_t$ to have density $h/w(t)$ with respect to Q_t, so that

$$\tilde{Q}_t(B) = \int_B \frac{h(x)}{w(t)}\, dQ_t(x).$$

(On the null set $w(t) = 0$, $\tilde{Q}_t$ can be an arbitrary probability measure.) Then

$$\begin{aligned} E_\theta f(X,T) &= E^* f(X,T) g_\theta(T) h(X) \\ &= \iint f(x,t) g_\theta(t) h(x)\, dQ_t(x)\, dG^*(t) \\ &= \iint f(x,t) \frac{h(x)}{w(t)}\, dQ_t(x) g_\theta(t) w(t)\, dG^*(t) \\ &= \iint f(x,t)\, d\tilde{Q}_t(x)\, dG_\theta(t). \end{aligned}$$

By (6.3) this shows that $\tilde{Q}$ is a conditional distribution for X given T under P_θ. Because $\tilde{Q}$ does not depend on θ, T is sufficient.

Before considering the converse—that if T is sufficient the densities p_θ must have form (6.7)—we should discuss mixture distributions. Given a marginal probability distribution G^* and a conditional distribution Q, we can define a *mixture distribution* $\hat{P}$ by

$$\hat{P}(B) = \int Q_t(B)\, dG^*(t) = \iint 1_B(x)\, dQ_t(x)\, dG^*(t).$$

---

[3] "Equivalence" here means that $\mu(N) = 0$ if and only if $P_\theta(N)$, $\forall \theta \in \Omega$. The assertion here is based on a result that any dominated family is equivalent to the mixture of some countable subfamily.

Then for integrable f,

$$\int f\,d\hat{P} = \iint f(x)\,dQ_t(x)\,dG^*(t).$$

(By linearity, this must hold for simple functions f, and the general case follows taking simple functions converging to f.)

Suppose now that T is sufficient, with Q the conditional distribution for X given T. Let g_θ be the G^* density of T when $X \sim P_\theta$. (This density will exist, for if $G^*(N) = 0$, $\mu(N_0) = 0$ where $N_0 = T^{-1}(N)$, and so $G_\theta(N) = P_\theta(T \in N) = P_\theta(X \in N_0) = \int_{N_0} p_\theta\,d\mu = 0$.) Then

$$\begin{aligned}
P_\theta(X \in B) &= E_\theta P_\theta(X \in B|T)\\
&= E_\theta Q_T(B)\\
&= \int Q_t(B) g_\theta(t)\,dG^*(t)\\
&= \iint 1_B(x)\,dQ_t(x)\,g_\theta(t)\,dG^*(t)\\
&= \iint 1_B(x) g_\theta\bigl(T(x)\bigr)\,dQ_t(x)\,dG^*(t)\\
&= \int_B g_\theta\bigl(T(x)\bigr)\,d\hat{P}(x).
\end{aligned}$$

This shows that P_θ has density $g_\theta\bigl(T(\cdot)\bigr)$ with respect to $\hat{P}$.

The mixture distribution $\hat{P}$ is absolutely continuous with respect to μ. To see this, suppose $\mu(N) = 0$. Then $P_\theta(N) = \int Q_t(N)\,dG_\theta(t) = 0$, which implies $G_\theta(\tilde{N}) = 0$, where $\tilde{N} = \{t : Q_t(N) > 0\}$. Because μ is equivalent to $\mathcal{P}$ and $G_\theta(\tilde{N}) = P_\theta(T \in \tilde{N}) = 0, \forall \theta \in \Omega$, $P^*(T \in \tilde{N}) = G^*(\tilde{N}) = 0$. Thus $Q_t(N) = 0$ (a.e. G^*) and so $\hat{P}(N) = \int Q_t(N)\,dG^*(t) = 0$. Taking $h = d\hat{P}/dP^*$, P_θ has density $g_\theta\bigl(T(x)\bigr)h(x)$ with respect to P^*.

6.5 Problems[4]

1. *The beta distribution.*
   a) Let X and Y be independent random variables with $X \sim \Gamma(\alpha, 1)$ and $Y \sim \Gamma(\beta, 1)$. Define new random variables $U = X + Y$ and $V = X/(X + Y)$. Find the joint density of U and V. Hint: If p is the joint density of X and Y, then

$$\begin{aligned}
P\{(U,V) \in B\} &= P\left\{\left(X+Y, \frac{X}{X+Y}\right) \in B\right\}\\
&= \int 1_B\left(x+y, \frac{x}{x+y}\right) p(x,y)\,dx\,dy.
\end{aligned}$$

[4] Solutions to the starred problems are given at the back of the book.

Next, change variables to write this integral as an integral against $u = x+y$ and $v = x/(x+y)$. The change of variables can be accomplished either using Jacobians or writing the double integral as an iterated integral and using ordinary calculus.

b) Find the marginal density for V. Use the fact that this density integrates to one to compute $\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\,dx$. This density for V is called the beta density with parameters α and β. The corresponding distribution is denoted $\mathrm{Beta}(\alpha, \beta)$.

c) Compute the mean and variance of the beta distribution.

d) Find the marginal density for U.

*2. Let X and Y be independent random variables with cumulative distribution functions F_X and F_Y.

a) Assuming Y is continuous, use smoothing to derive a formula expressing the cumulative distribution function of X^2Y^2 as the expected value of a suitable function of X. Also, if Y is absolutely continuous, give a formula for the density.

b) Suppose X and Y are both exponential with the same failure rate λ. Find the density of $X - Y$.

*3. Suppose that X and Y are independent and positive. Use a smoothing argument to show that if $x \in (0,1)$, then

$$P\left(\frac{X}{X+Y} \le x\right) = EF_X\left(\frac{xY}{1-x}\right), \tag{6.8}$$

where F_X is the cumulative distribution function of X.

*4. Differentiating (6.8), if X is absolutely continuous with density p_X, then $V = X/(X+Y)$ is absolutely continuous with density

$$p_V(x) = E\left[\frac{Y}{(1-x)^2} p_X\left(\frac{xY}{1-x}\right)\right], \qquad x \in (0,1).$$

Use this formula to derive the beta distribution introduced in Problem 6.1, showing that if X and Y are independent with $X \sim \Gamma(\alpha, 1)$ and $Y \sim \Gamma(\beta, 1)$, then $V = X/(X+Y)$ has density

$$p_V(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$$

for $x \in (0,1)$.

*5. Let X and Y be absolutely continuous with joint density

$$p(x,y) = \begin{cases} 2, & 0 < x < y < 1; \\ 0, & \text{otherwise.} \end{cases}$$

a) Find the marginal density of X and the marginal density of Y.

b) Find the conditional density of Y given $X = x$.

c) Find $E[Y|X]$.
d) Find EXY by integration against the joint density of X and Y.
e) Find EXY by smoothing, using the conditional expectation you found in part (c).

*6. Let μ be Lebesgue measure on $\mathbb{R}$ and let ν be counting measure on $\{0,1,\ldots\}^2$. Suppose the joint density of X and Y with respect to $\mu \times \nu$ is given by

$$p(x, y_1, y_2) = x^2(1-x)^{y_1+y_2}$$

for $x \in (0,1)$, $y_1 = 0,1,2,\ldots$, and $y_2 = 0,1,2,\ldots$.
a) Find the marginal density of X.
b) Find the conditional density of X given $Y = y$ (i.e., given $Y_1 = y_1$ and $Y_2 = y_2$).
c) Find $E[X|Y]$ and $E[X^2|Y]$. Hint: The formula

$$\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\,dx = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

may be useful.
d) Find $E\big[1/(4+Y_1+Y_2)\big]$. Hint: Find EX using the density in part (a) and find an expression for EX using smoothing and the conditional expectation in part (c).

7. Let X and Y be random variables with joint Lebesgue density

$$p(x,y) = \begin{cases} 2y^2e^{-xy}, & x > 0,\ y \in (0,1); \\ 0, & \text{otherwise.} \end{cases}$$

a) Find the marginal density for Y.
b) Find the conditional density for X given $Y = y$.
c) Find $P(X > 1|Y = y)$, $E[X|Y = y]$, and $E[X^2|Y = y]$.

8. Suppose X has the standard exponential distribution with marginal density e^{-x}, $x > 0$, and that

$$P(Y = y|X = x) = \frac{x^y e^{-x}}{y!}, \qquad y = 0,1,\ldots.$$

a) Find the joint density for X and Y. Identify the dominating measure.
b) Find the marginal density of Y.
c) Find the conditional density of X given $Y = y$.
d) Find EY using the marginal density in part (b).
e) As the conditional distribution of Y given $X = x$ is Poisson with parameter x, $E(Y|X = x) = x$. Use this to find EY by smoothing.

9. Suppose that X is uniformly distributed on the interval $(0,1)$ and that

$$P(Y = y|X = x) = (1-x)x^y, \qquad y = 0,1,\ldots.$$

a) Find the joint density for X and Y. What is the measure for integrals against this density?

b) Find the marginal density of Y.
c) Find the conditional density of X given $Y = y$.
d) Find $E[X|Y = y]$. Find $P(X < 1/2|Y = 0)$ and $P(X < 1/2|Y = 1)$.

10. Suppose X and Y are independent, both uniformly distributed on $(0, 1)$. Let $M = \max\{X, Y\}$ and $Z = \min\{X, Y\}$.
  a) Find the conditional distribution of Z given $M = m$.
  b) Suppose instead that X and Y are independent but uniformly distributed on the finite set $\{1, \dots, k\}$. Give the conditional distribution of Z given $M = j$.

*11. Suppose X and Y are independent, both absolutely continuous with common density f. Let $M = \max\{X, Y\}$ and $Z = \min\{X, Y\}$. Determine the conditional distribution for the pair (X, Y) given (M, Z).

*12. Let X and Y be independent exponential variables with failure rate λ, so the common marginal density is $\lambda e^{-\lambda x}$, $x > 0$. Let $T = X + Y$. Give a formula expressing $E[f(X, Y)|T = t]$ as a one-dimensional integral. Hint: Review the initial example on sufficiency in Section 3.2.

13. Suppose X and Y are absolutely continuous with joint density

$$\frac{\exp\left[-\frac{x^2-2\rho xy+y^2}{2(1-\rho^2)}\right]}{2\pi\sqrt{1-\rho^2}}.$$

This is a bivariate normal density with

$$EX = EY = 0, \qquad \mathrm{Var}(X) = \mathrm{Var}(Y) = 1,$$

and

$$\mathrm{Cor}(X, Y) = \rho.$$

Determine the conditional distribution of Y given X. (Naturally, the answer should depend on the correlation coefficient ρ.) Use smoothing to find the covariance between X^2 and Y^2.

*14. Let X and Y be absolutely continuous with density $p(x, y) = e^{-x}$, if $0 < y < x$; $p(x, y) = 0$, otherwise.
  a) Find the marginal densities of X and Y.
  b) Compute EY and EY^2 integrating against the marginal density of Y.
  c) Find the conditional density of Y given $X = x$, and use it to compute $E[Y|X]$ and $E[Y^2|X]$.
  d) Find the expectations of $E[Y|X]$ and $E[Y^2|X]$ integrating against the marginal density of X.

15. Suppose X has a Poisson distribution with mean λ and that given $X = x$, Y has a binomial distribution with x trials and success probability p. (If $X = 0$, $Y = 0$.)
  a) Find the marginal distribution of Y.
  b) Find the conditional distribution of X given Y.
  c) Find $E[Y^2|X]$.
  d) Compute EY^2 by smoothing, using the result in part (c).

e) Compute EY^2 integrating against the marginal distribution from part (a).
f) Find $E[X|Y]$ and use this to compute EX by smoothing.

16. Let $X = (X_1, X_2)$ be an absolutely continuous random vector in $\mathbb{R}^2$ with density f, and let $T = X_1 + X_2$.
a) Find the joint density for X_1 and T.
b) Give a formula for the density of T.
c) Give a formula for the conditional density of X_1 given $T = t$.
d) Give a formula for $E\big[g(X) \bigm| T = t\big]$. Hint: View $g(X)$ as a function of T and X_1 and use the conditional density you found in part (c).
e) Suppose X_1 and X_2 are i.i.d. standard normal. Then $X_1 - X_2 \sim N(0, 2)$ and $T \sim N(0, 2)$. Find $P\big(|X_1 - X_2| < 1 \bigm| T\big)$ using your formula from part (d). Integrate this against the density for T to show that smoothing gives the correct answer.

17. Let $X_1, \dots, X_n$ be jointly distributed Bernoulli variables with mass function

$$P(X_1 = x_1, \dots, X_n = x_n) = \frac{s_n!(n - s_n)!}{(n+1)!},$$

where $s_n = x_1 + \cdots + x_n$.
a) Find the joint mass function for $X_1, \dots, X_{n-1}$. (Your answer should simplify.)
b) Find the joint mass function for $X_1, \dots, X_k$ for any $k < n$.
c) Find $P(X_{k+1} = 1 | X_1 = x_1, \dots, X_k = x_k)$, $k < n$.
d) Let $S_n = X_1 + \cdots + X_n$. Find

$$P(X_1 = x_1, \dots, X_n = x_n | S_n = s).$$

e) Let $Y_k = (1 + S_k)/(k + 2)$. For $k < n$, find

$$E(Y_{k+1} | X_1 = x_1, \dots, X_k = x_k),$$

expressing your answer as a function of Y_k. Use smoothing to relate EY_{k+1} to EY_k. Find EY_k and ES_k.

18. Suppose $X \sim N(0, 1)$ and $Y | X = x \sim N(x, 1)$.
a) Find the mean and variance of Y.
b) Find the conditional distribution of X given $Y = y$.

19. Let X be absolutely continuous with a positive continuous density f and cumulative distribution function F. Take $Y = X^2$.
a) Find the cumulative distribution function and the density for Y.
b) For $y > 0$, $y \neq x^2$, find

$$\lim_{\epsilon \downarrow 0} P\big[X \le x \bigm| Y \in (y - \epsilon, y + \epsilon)\big].$$

c) The limit in part (b) should agree with the cumulative distribution function for a discrete probability measure Q_y. Find the mass function for this discrete distribution.

d) Show that Q is a conditional distribution for X given Y. Specifically, show that it satisfies the conditions in Definition 6.2.

20. Suppose X and Y are conditionally independent given $W = w$ with

$$X|W = w \sim N(aw, 1) \text{ and } Y|W = w \sim N(bw, 1).$$

Use smoothing to derive formulas relating EX, EY, $\mathrm{Var}(X)$, $\mathrm{Var}(Y)$, and $\mathrm{Cov}(X, Y)$ to moments of W and the constants a and b.

21. Suppose $Y \sim N(\nu, \tau^2)$ and that given $Y = y$, $X_1, \dots, X_n$ are i.i.d. from $N(y, \sigma^2)$. Taking $\overline{x} = (x_1 + \cdots + x_n)/n$, show that the conditional distribution of Y given $X_1 = x_1, \dots, X_n = x_n$ is normal with

$$E(Y|X_1 = x_1, \dots, X_n = x_n) = \frac{\nu/\tau^2 + n\overline{x}/\sigma^2}{1/\tau^2 + n/\sigma^2}$$

and

$$\mathrm{Var}(Y|X_1 = x_1, \dots, X_n = x_n) = \frac{1}{1/\tau^2 + n/\sigma^2}.$$

Remark: If *precision* is defined as the reciprocal of the variance, these formulas state that the precision of the conditional distribution is the sum of the precisions of the X_i and Y, and the mean of the conditional distribution is an average of the X_i and ν, weighted by the precisions of the variables.

22. A building has a single elevator. Times between stops on the first floor are presumed to follow an exponential distribution with failure rate θ. In a time interval of duration t, the number of people who arrive to ride the elevator has a Poisson distribution with mean λt.
    a) Suggest a joint density for the time T between elevator stops and the number of people X that board when it arrives.
    b) Find the marginal mass function for X.
    c) Find EX^2.
    d) Let $\lambda > 0$ and $\theta > 0$ be unknown parameters, and suppose we observe data $X_1, \dots, X_n$ that are i.i.d. with the marginal mass function of X in part (b). Suggest an estimator for the ratio θ/λ based on the average $\overline{X}$. With these data, if n is large will we be able to estimate λ accurately?